

FEDERATED LEARNING PRIVACY AND SECURITY ENHANCEMENT

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Abstract—Federated Learning (FL) has emerged as a promising alternative to centralized machine learning by enabling collaborative model training without direct data sharing. With recent advances in FL on feature analysis, aggregation algorithms, privacy, and security, FL improves data locality and lowers raw data exposure compared to centralized learning. However, it introduces new methods of attacks such as model poisoning, inference/membership attacks and important tradeoffs among communication cost, robustness, and privacy guarantees. We compare classical techniques such as federated averaging with robust aggregation strategies including Trimmed Mean, Krum, Bulyan, and Robust Federated Aggregation, highlighting trade-offs in computational efficiency, scalability, and resilience against compromised clients. Our analysis shows how FL improves data privacy through secure aggregation and differential privacy, and how it deals with threats such as poisoning and inference attacks. A survey of recent patents shows developing interest in communication efficiency, adaptive aggregation and privacy solutions. Overall, FL demonstrates strong potential for privacy-first machine learning, but faces challenges in robustness, communication overhead, and data heterogeneity that define areas of interest for future research

I. RESULTS

A. Average Loss

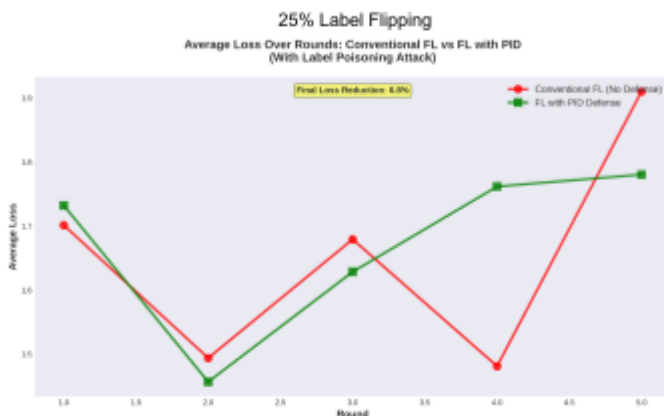


Fig. 1. Average loss per round in Conventional FL vs FL with PID with 25% label flipping

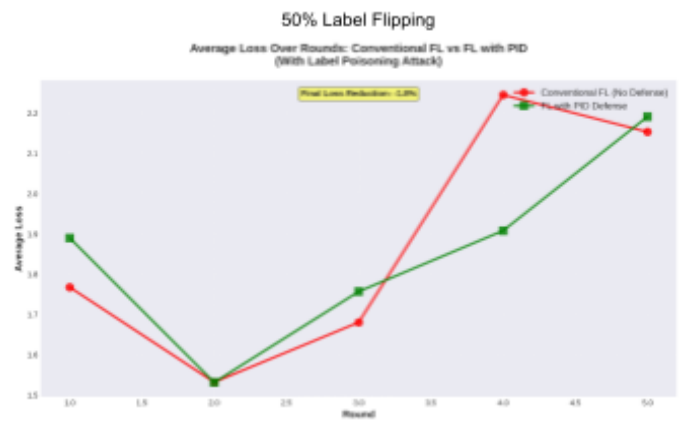


Fig. 2. Average loss per round in Conventional FL vs FL with PID with 50% label flipping

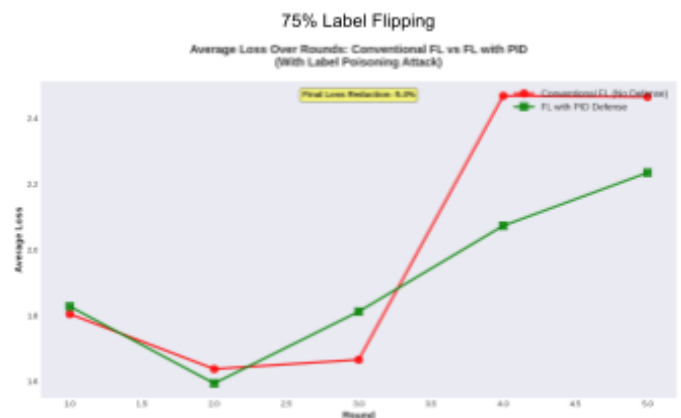


Fig. 3. Average loss per round in Conventional FL vs FL with PID with 75% label flipping

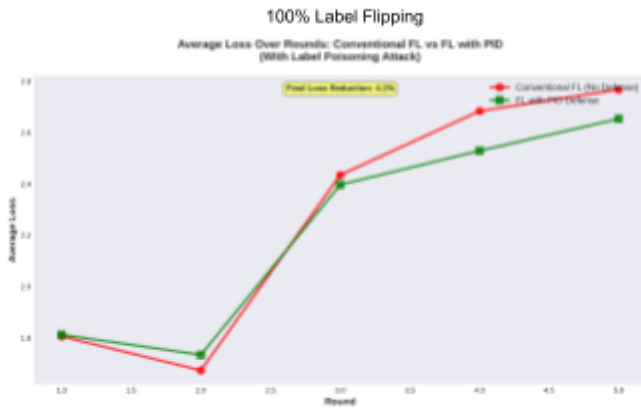


Fig. 4. Average loss per round in Conventional FL vs FL with PID with 100% label flipping

As shown in the Figures above (Fig. 1-4) Federated Learning with PID tends to cause the average loss per round to decrease faster compared to conventional FL. Although in Figure 2 with 50% label flipping the loss for FL with PID is higher than conventional FL, this is most likely due to the fact that there were only 2 poisoned clients and only 5 rounds of FL which could have resulted in some rounds to have excluded the poisoned clients during training or testing.



Fig. 5. Average accuracy per round in Conventional FL vs FL with PID with 25% label flipping

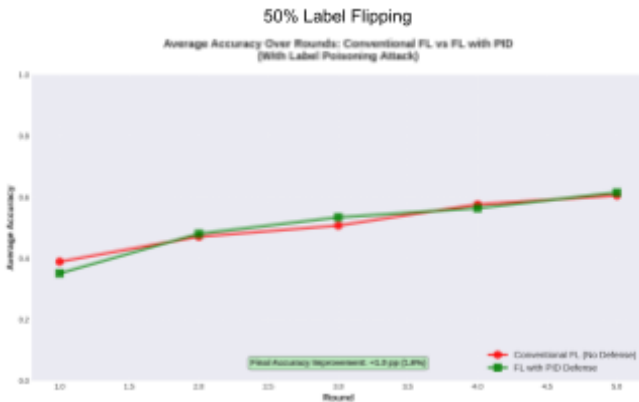


Fig. 6. Average accuracy per round in Conventional FL vs FL with PID with 50% label flipping

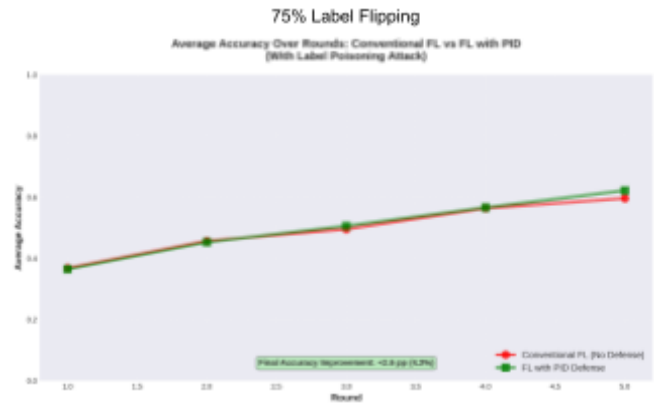


Fig. 7. Average accuracy per round in Conventional FL vs FL with PID with 75% label flipping

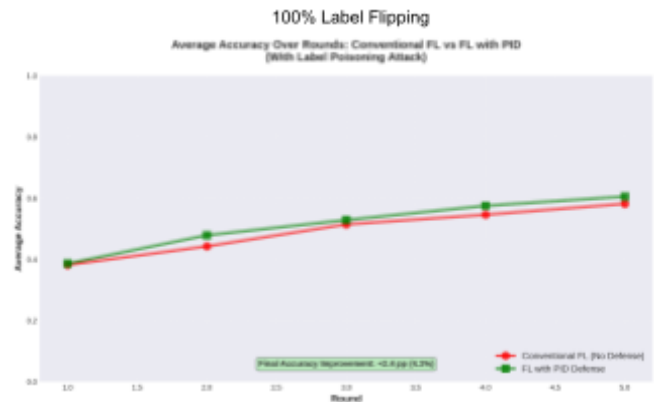


Fig. 8. Average accuracy per round in Conventional FL vs FL with PID with 100% label flipping

B. Average Accuracy

Similarly with the average accuracy per round FL with PID tends to have the higher accuracy compared to conventional FL and in 50% label flipping the conventional FL may have been lucky enough to avoid training or testing on the poisoned clients. However, as shown in the other Figures (Fig. 5, 7, 8) FL with PID produces a higher accuracy almost consistently.

C. Impact of Attack Severity

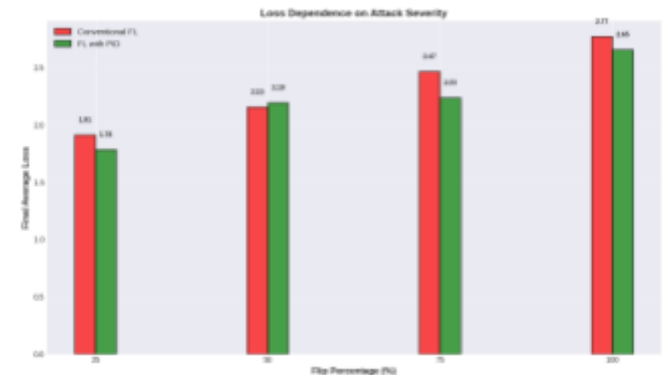


Fig. 9. Impact of attack severity (label flipping) on the average loss of the final round.

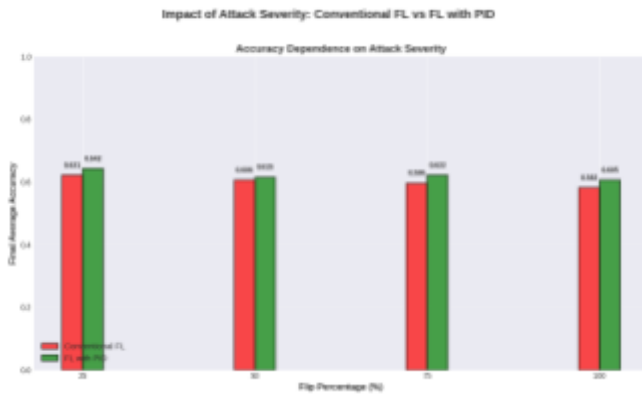


Fig. 10. Impact of attack severity (label flipping) on the average accuracy of the final round.

As shown in Figure 9 the Final average loss tends to be higher in conventional FL increasing with the attack severity. Finally, in Figure 10, the final average accuracy for conventional FL decreases from 62% to 58% as the attack severity increases and is lower compared to FL with PID. However, in FL with the PID, the accuracy barely changes, staying around 60% accuracy which shows that the PID did protect itself from the poisoned clients and maintained a standard accuracy.

II. CONCLUSION

From the results above, we can conclude that FL with PID does in fact work and block out poisoned clients, providing more stability and faster convergence. However, the hurdle with PID is adjusting the coefficients in the PID formula to obtain the maximum effect of the PID. The optimal implementation would be a dynamic implementation where the coefficients can evolve with the data it is given, however, it would have to be able to detect if data is evolving or poisoned.

Appendix: A. Average loss per round in Conventional FL vs FL with PID

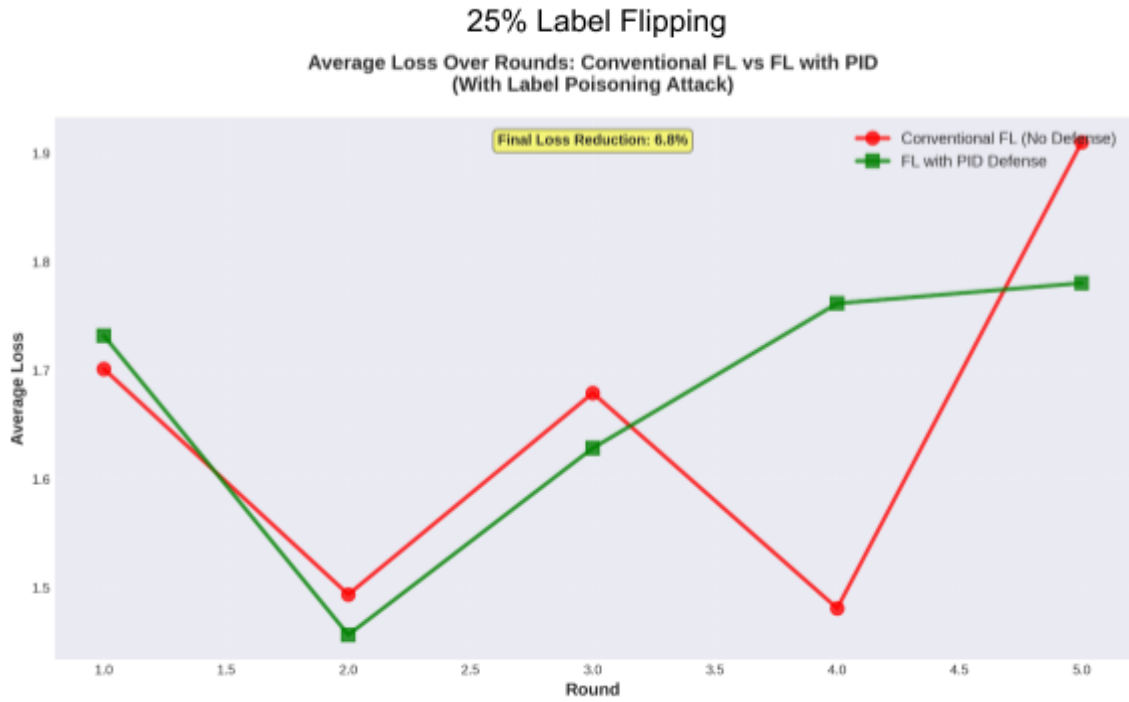


Fig. 1. Average loss per round in Conventional FL vs FL with PID with 25% label flipping

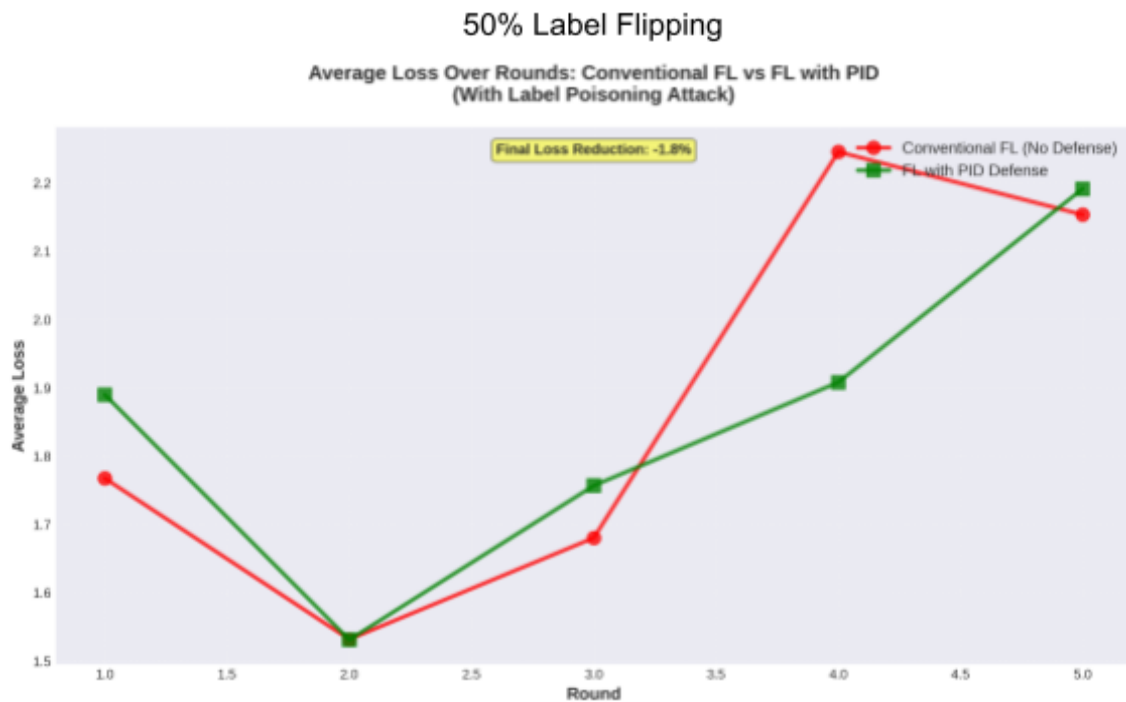


Fig. 2. Average loss per round in Conventional FL vs FL with PID with 50% label flipping

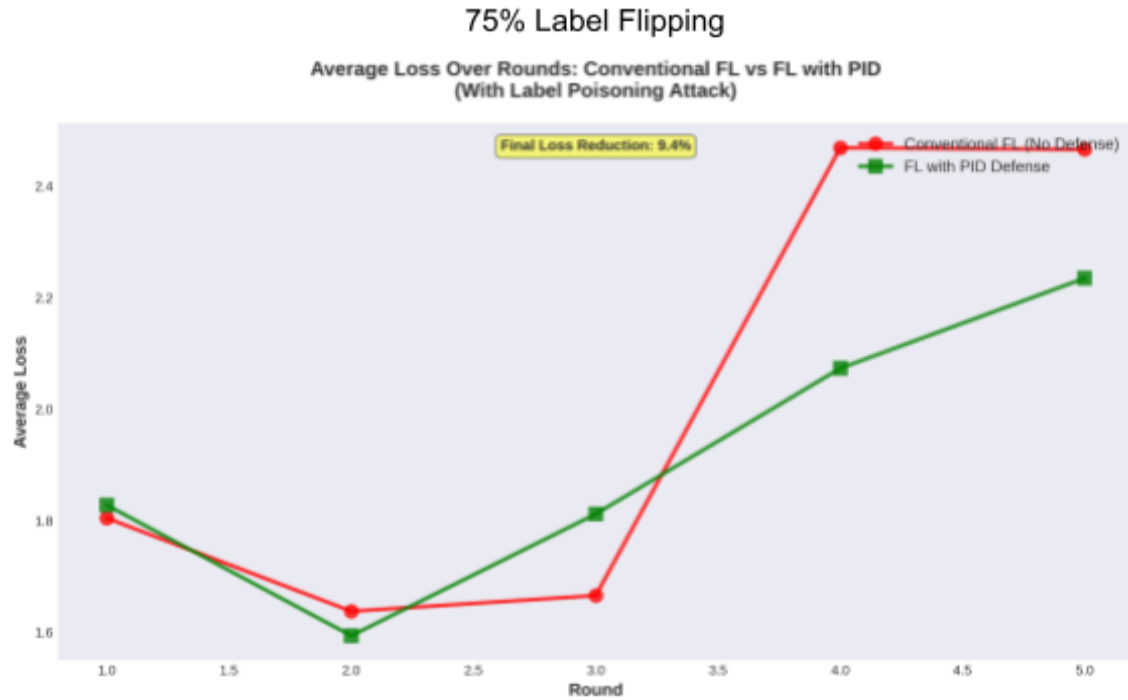


Fig. 3. Average loss per round in Conventional FL vs FL with PID with 75% label flipping

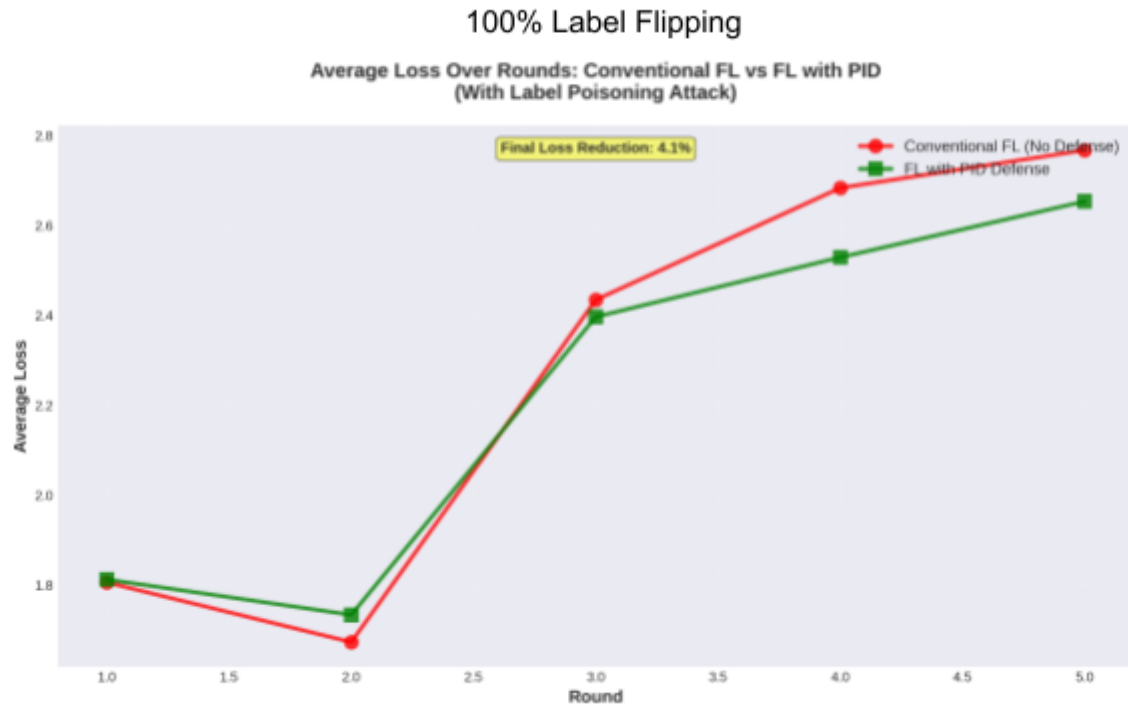


Fig. 4. Average loss per round in Conventional FL vs FL with PID with 100% label flipping

Appendix: B. Average Accuracy per round in Conventional FL vs FL with PID

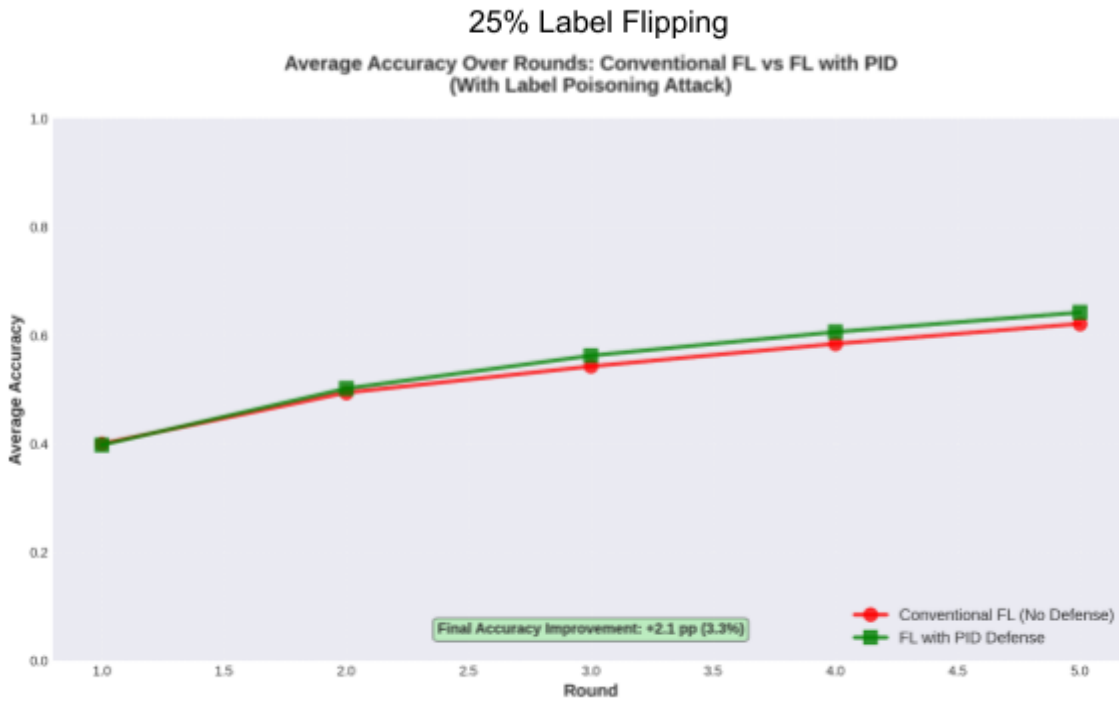


Fig. 5. Average accuracy per round in Conventional FL vs FL with PID with 25% label flipping

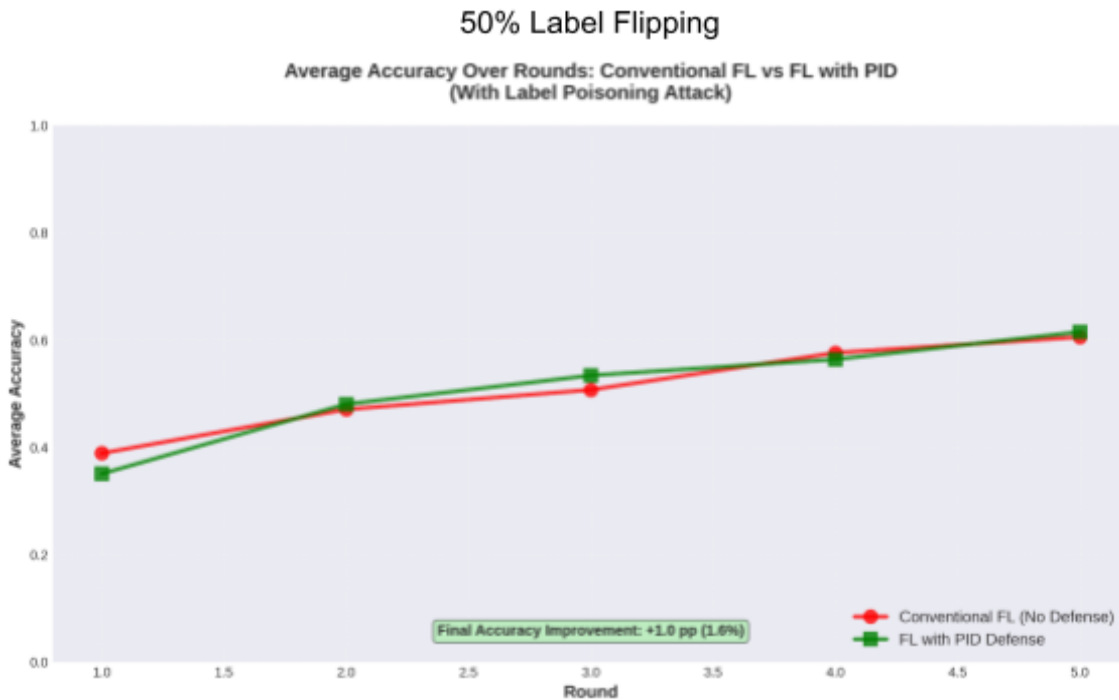


Fig. 6. Average accuracy per round in Conventional FL vs FL with PID with 25% label flipping

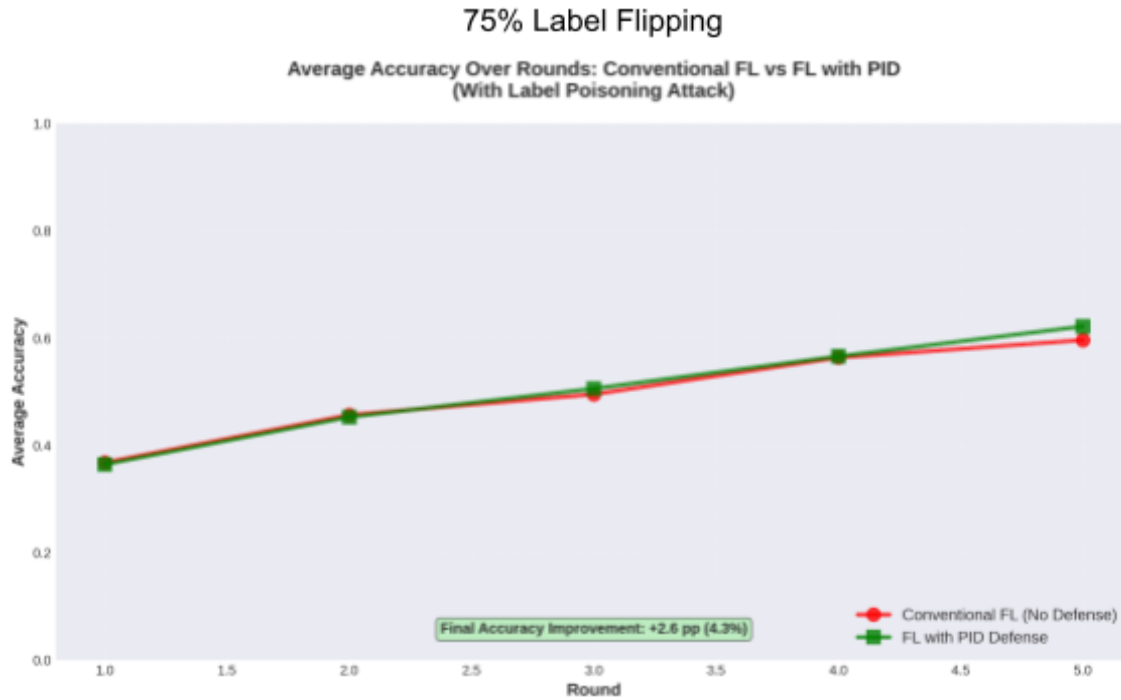


Fig. 7. Average accuracy per round in Conventional FL vs FL with PID with 75% label flipping

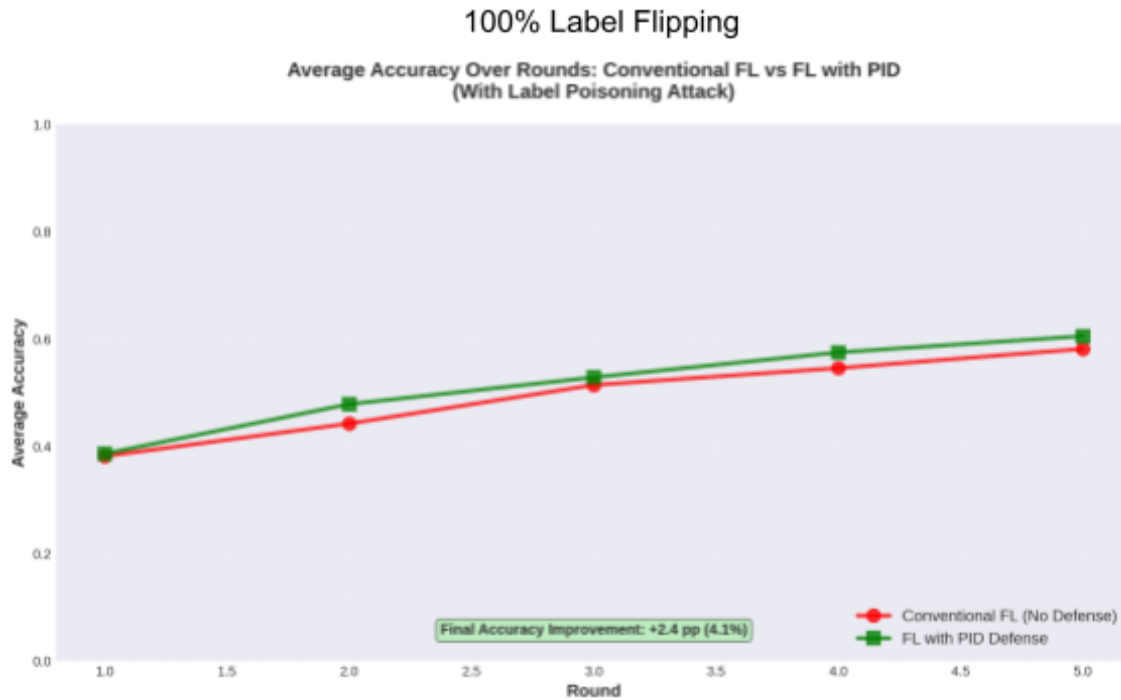


Fig. 8. Average accuracy per round in Conventional FL vs FL with PID with 100% label flipping

Appendix: C. Impact of Attack Severity in Conventional FL vs FL with PID

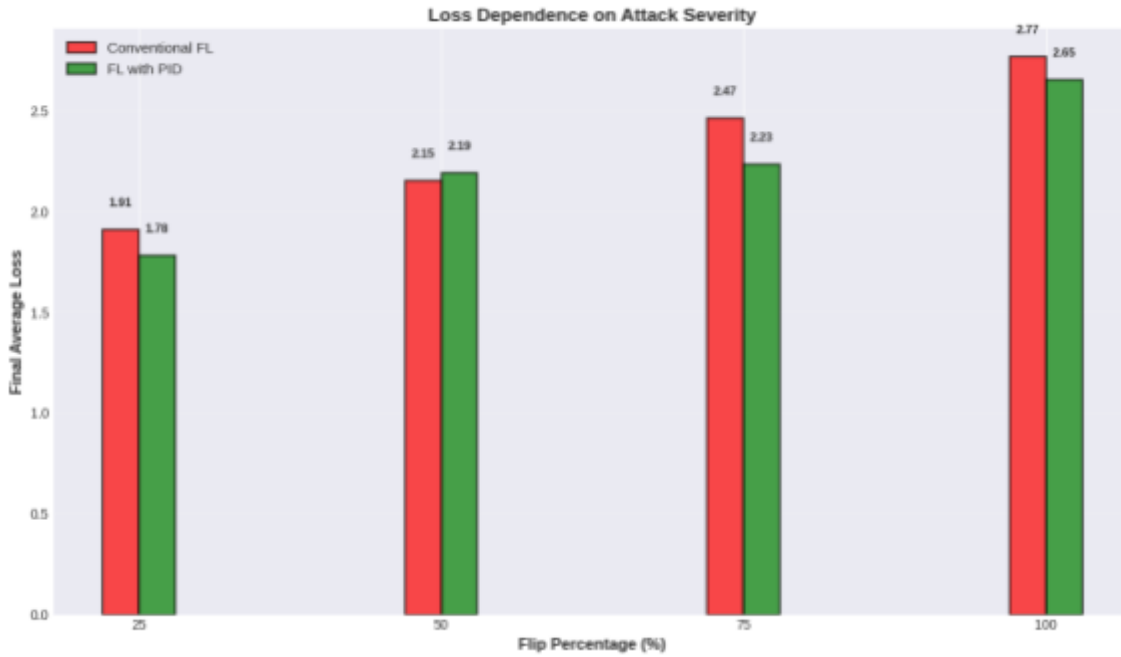


Fig. 9. Impact of attack severity (label flipping) on the average loss of the final round.

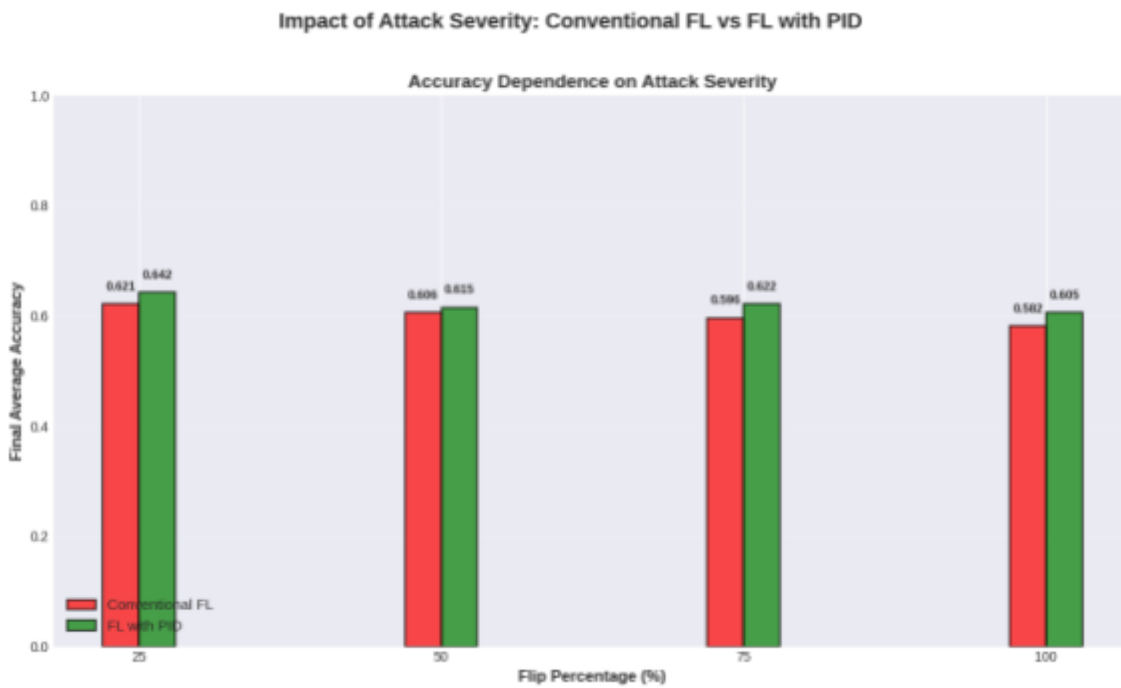


Fig. 11. Impact of attack severity (label flipping) on the average accuracy of the final round.